

Screening Questions (maximum Of 2.000 Characters Per Question)

- Part of this job will require annual reporting to the Scientific Committee of the WCPFC. Briefly describe your experience preparing scientific reports, the tools you employ for efficiency and repeatability, statistical approaches you have used, and key considerations for data visualizations.

I have extensive experience preparing scientific reports for international organisations. As Postdoctoral Scientist for the IUCN Shark Specialist Group, I coordinated 60+ scientists to develop and publish technical documentation, ensuring consistency and quality across contributions. Over my career, I have also contributed to five scientific reports for IUCN and the Government of New Zealand, and authored 12 peer-reviewed articles in journals including *Nature Communications*, *Global Change Biology*, and *Marine Policy*.

For efficiency and repeatability, I use R and Quarto to create reproducible analytical pipelines where data processing, analysis, and reporting are integrated into a single workflow. This approach minimises manual tasks and ensures outputs can be updated efficiently when new data become available. Version control with Git ensures transparency and allows collaborators to track changes. I have also built Shiny applications that automate data validation and visualization generation, reducing repetitive work and enabling team members to focus on higher-value analytical tasks.

My statistical expertise spans inferential and predictive modelling across large-scale datasets, including spatial analyses and machine learning approaches.

For data visualizations, I excel in designing advanced and complex charts with ggplot2, tailoring visualizations to effectively communicate patterns in large datasets. I prioritise clarity and accessibility: choosing appropriate chart types, using colourblind-friendly palettes, and ensuring figures can stand alone with clear labels and legends. This is essential when communicating findings to diverse stakeholders.

- Fisheries management relies on accurate and representative data to inform the scientific work for decision making. Errors in fisheries data can be cryptic and contextual. Briefly describe the approaches you would employ to validate and ensure the quality of fisheries data and provide an example, from a past role, that required a deep dive into data to uncover issues.

Ensuring data quality requires a combination of automated checks, domain expertise, and close collaboration with data providers. My approach involves several layers:

- First, I implement systematic validation routines: checking for missing values, outliers, duplicates, and logical inconsistencies (e.g., catch weights exceeding vessel capacity, impossible dates, coordinates outside expected ranges). These checks are scripted in R to ensure repeatability and can flag records for review without manual inspection of every entry.

- Second, I cross-reference data against external sources where possible—comparing reported values with independent estimates, validating taxonomic names against authoritative databases, or checking spatial data against known boundaries. Contextual errors often emerge only when comparing across datasets or time periods.
- Third, I explore data visually and statistically to identify anomalies: mapping spatial distributions, plotting time series, and examining relationships between variables. Unexpected patterns often reveal data entry errors, changes in reporting protocols, or sampling biases. Crucially, resolving these issues requires dialogue with data providers to understand the context behind anomalies rather than making assumptions.

An example: while developing the FINS Shiny application for fossil shark data compiled from multiple institutions worldwide, I discovered systematic inconsistencies in taxonomic classifications across records. Some species names were outdated synonyms, others misspelled, and certain records used inconsistent hierarchies. I implemented a validation module that cross-references entries against a curated taxonomic backbone, flags mismatches, and suggests corrections. This revealed that approximately 15% of records required revision. These errors would have propagated into downstream diversity analyses if undetected. The tool now enables collaborators to validate and correct their own contributions efficiently.

- In situations where data collection is limited, we must make assumptions to best estimate tuna catches. Please describe possible approaches to use limited sampling from port sampling or logbook data to estimate total catch removals from a broader area and highlight the key uncertainties associated.

When sampling coverage is limited, estimation approaches must balance statistical rigour with practical constraints while clearly communicating uncertainties to decision-makers. Several methods can be employed:

Ratio estimators: Using sampled catch-per-unit-effort (CPUE) multiplied by total effort to estimate total catch. This assumes sampled vessels or trips are representative of the broader fleet.

Stratified expansion: Dividing the fishery into strata (by area, time period, gear type, or vessel class) and applying stratum-specific raising factors. If port sampling covers 20% of landings in a stratum, catches are raised accordingly. Stratification reduces bias when catch rates vary across groups.

Model-based approaches: Fitting generalised linear or additive models to predict catch based on covariates such as location, season, effort, and environmental conditions. Models can interpolate to unsampled areas or periods, provided covariates are available.

Spatial interpolation: For spatially referenced data, geostatistical methods can estimate catches in unsampled areas based on spatial autocorrelation patterns.

Key uncertainties include:

- **Non-random sampling:** If sampled vessels or ports differ systematically from the broader fleet, estimates will be biased.
- **Coverage gaps:** Estimates for unsampled strata rely on assumptions about similarity with sampled strata.
- **Reporting accuracy:** Logbook data may contain misreporting, species misidentification, or underreporting of bycatch.
- **Effort estimation:** Total effort is often uncertain, and errors propagate into catch estimates.

Quantifying these uncertainties through simulation or resampling approaches is essential, and working closely with member countries to understand local sampling contexts helps identify where assumptions may be weakest.